MILP formulation Partitioning and Scheduling

1 Scheduling Constrained Partitioning Problem with MILP

1.1 Problem Definition

We are given a graph $G = (V, E)$ with n nodes and m edges. Each node has weights h_i and s_i to denote hardware and software execution times. Also, each node has an associated hardware area cost a_i. The edges denote communication between the nodes and have associated communication cost c_{ij} for the edge e connecting nodes i and j. The communication delay is taken into account only when the adjacent nodes belong to different partitions. The aim is to partition the nodes V into non-overlapping sets V_S and V_H denoting software and hardware nodes respectively; $V = V_H \cup V_S$ and $V_H \cap V_S = \emptyset$. We denote the set of edges between the two sets by $E_{\mathcal{P}}$. Additionally, we need to ensure that software nodes are executed sequentially, while hardware nodes can execute in parallel if their dependent nodes are executed. Our objective is to minimize the overall execution time given a limited available area for hardware nodes.

1.2 Optimization Formulation

Decision variables. We define binary variable $x_i = 0$ if node i is assigned to hardware partition, $x_i = 1$ if assigned to software partition. We also define binary variable z_{ij} for each edge $e := (i, j) \in E$connecting nodes i and j. We denote $z_{ij} = 1$ if the nodes i and j belong to different partitions. Let t_i and f_i denote the start and end time of the node i. The total execution time of the task graph is denoted by T. We also define binary variable y_{ij} for each pair of nodes i, j, $i \neq j$. This binary variable is used to ensure that if the pair of nodes i, j are assigned to software partition, they need to be executed sequentially; i.e., if $y_{ij} = 1$, then node i precedes node j, and for $y_{ij} = 0$, the opposite happens. However, we need to ensure that this sequential constraint is activated for only a pair of software nodes.

Hardware area constraint. The overall hardware area is constrained by the maximum allowable hardware area $A_{\max}$.

$$\sum_{i=1}^{n} a_i \,(1 - x_i) \leq A_{\max} \tag{1}$$

Binary edge variable. The binary variable z_{ij} and the partition variable x_i, x_j corresponding to the adjacent nodes are related by

$$z_{ij} \geq x_i - x_j, \qquad \forall\,(i, j) \in E \tag{2a}$$

$$z_{ij} \geq x_j - x_i, \qquad \forall\,(i, j) \in E \tag{2b}$$

$$z_{ij} \leq x_i + x_j, \qquad \forall\,(i, j) \in E \tag{2c}$$

$$z_{ij} \leq 2 - x_i - x_j, \qquad \forall\,(i, j) \in E \tag{2d}$$

Execution time constraints. The execution time of each node is denoted

$$
\begin{aligned}
& T \geq f_i, && \forall i \in V && \text{(3a)}\\
& f_i = t_i + h_i \left(1 - x_i\right) + s_i x_i, && \forall i \in V && \text{(3b)}\\
& f_i + c_{ij} z_{ij} \leq t_j, && \forall \left(i, j\right) \in E && \text{(3c)}
\end{aligned}
$$

Sequential execution of software nodes. We define the following pair of constraints with M being a large positive number.

$$
\begin{aligned}
& f_i \leq t_j + M \left(1 - y_{ij}\right) + M \left(2 - x_i - x_j\right), && \forall i, j \in V, i \neq j && \text{(4a)}\\
& f_j \leq t_i + M y_{ij} + M \left(2 - x_i - x_j\right), && \forall i, j \in V, i \neq j && \text{(4b)}
\end{aligned}
$$

Note that, the constraints are activated only when $x_i = x_j = 1$ or the nodes i, j are both software nodes, when the term $2 - x_i - x_j = 0$. For other cases, the constraints are not active since M is sufficiently large. Once the constraints are activated, the value of y_{ij} denotes the precedence of the software node operations.

The overall optimization problem is

$$
\begin{aligned}
& \text{minimize } T && \text{(5a)}\\
& \text{subject to } \left(1\right), \left(2\right), \left(3\right), \left(4\right) && \text{(5b)}
\end{aligned}
$$

Note that the pairwise sequential execution constraints for software nodes scale quadratically, making the MILP formulation computationally expensive for large task graphs.